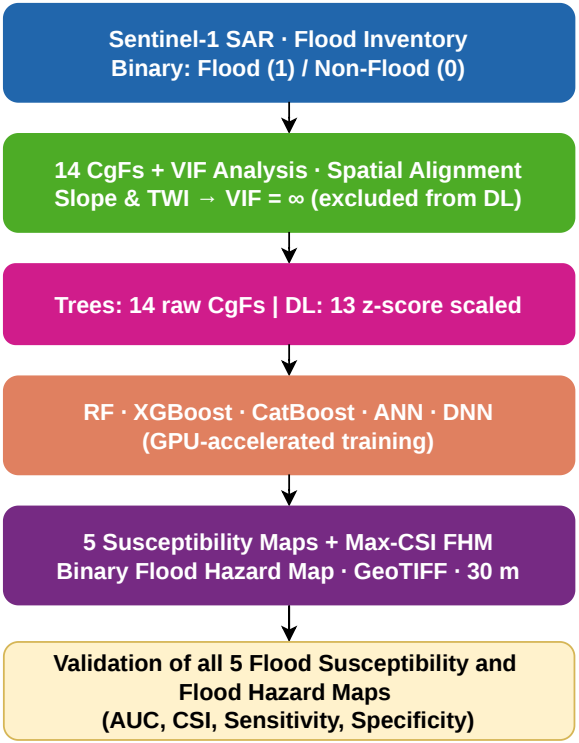

Model Performance

Model	AUC	CSI	Sensitivity	Specificity
XGBoost ★	**0.9982**	**0.937**	**0.967**	**0.988**
CatBoost	0.9935	0.871	0.930	0.974
Rand. Forest	0.9910	0.844	0.919	0.967
DNN	0.9615	0.705	0.911	0.890
ANN	0.9515	0.668	0.883	0.880

Threshold Comparison (Max-CSI vs Youden's Index)

Model	Max-CSI τ*	Youden τ	ΔCSI
XGBoost	0.491	0.483	+0.008
CatBoost	0.500	0.483	+0.017
RF	0.482	0.466	+0.016